

Supervised Contrastive Learning Leads to More Reasonable Spectral Embeddings

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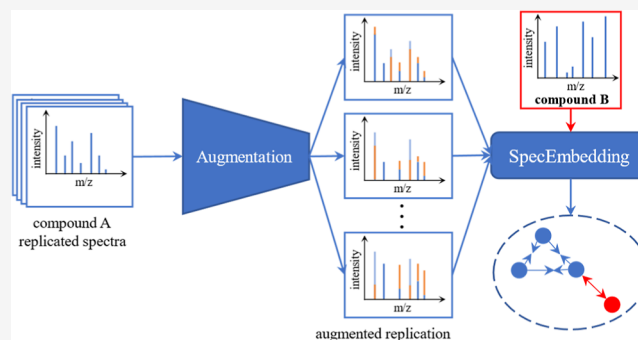


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ABSTRACT: Over the past decades, mass spectrometry has served as a fundamental technique for molecular identification in the field of metabolomics, widely applied to the analysis and characterization of biomolecules. However, the complexity of experimental conditions and the structural similarities among compounds pose significant challenges for accurate identification. Recent advancements in deep learning have offered new solutions to address these challenges, particularly demonstrating great potential in generating high-quality spectral embeddings. In this study, we propose a novel method, SpecEmbedding, which leverages a transformer encoder architecture and employs replicated spectra of compounds as positive samples, trained under a supervised contrastive learning framework. By mapping complex mass spectra into low-dimensional vector representations, SpecEmbedding significantly enhances the comparability between spectra, thereby improving identification accuracy. We trained SpecEmbedding on the GNPS training subset and evaluated its performance on the GNPS test subset, the MoNA data set, and the MTBLS1572 data set. Experimental results show that SpecEmbedding achieves a Top-1 hit ratio of 81.73% on the GNPS test subset, substantially outperforming MSBert (77.81%) and DreaMS (71.90%). The source code for this study is publicly available at <https://github.com/sword-nan/SpecEmbedding>. A web service is provided at <https://huggingface.co/spaces/xp113280/SpecEmbedding>.



INTRODUCTION

With the rapid advancement of mass spectrometry technologies, large-scale spectral databases such as GNPS,¹ HMDB,² and MassBank³ have accumulated hundreds of millions of mass spectra. In a single mass spectrometry experiment, tens of thousands of spectra are typically acquired from the instrument. Efficient and accurate analysis of this vast amount of data remains a critical challenge for researchers.⁴ In the compound identification process, a common approach is to rely on reference databases, where candidate compounds are selected based on the similarity between the experimental spectra and reference spectra,^{5,6} with those achieving the highest similarity scores being chosen as identification results. For decades, cosine similarity and its variants^{4,7,8} have been the predominant methods for similarity computation. However, due to their computational limitations and heavy reliance on high-quality spectral data, researchers have increasingly sought more efficient and robust alternatives.⁹

In recent years, emerging machine learning and deep learning techniques^{10,11} have demonstrated tremendous potential in the field of mass spectrum analysis. These approaches typically either directly predict molecular fingerprints or extract representation vectors from spectra for downstream analyses, enabling the effective capture of the complex relationships between compounds and their spectral

patterns, thereby significantly improving compound identification accuracy.^{12–14} Spec2Vec⁹ is an unsupervised learning method based on the word2vec¹⁵ model, where an entire mass spectrum is analogized to a document in natural language, and spectral peaks are treated as words. It learns peak representations by predicting peak contexts and ultimately obtains the spectrum embedding through a weighted combination of the individual peak embeddings. MS2DeepScore¹⁶ adopts a peak binning strategy to convert spectra into fixed-dimensional vectors, employs a Siamese network architecture,¹⁷ and maps them into a target embedding space via a multilayer perceptron (MLP), using the mean squared error (MSE) of the Tanimoto similarity as the loss function for training. DreaMS¹⁸ introduces Fourier encoding¹⁹ to characterize the mass-to-charge ratio (m/z) features and utilizes a Graph Transformer²⁰ as the core model. It is trained in a self-supervised manner through peak masking and elution order

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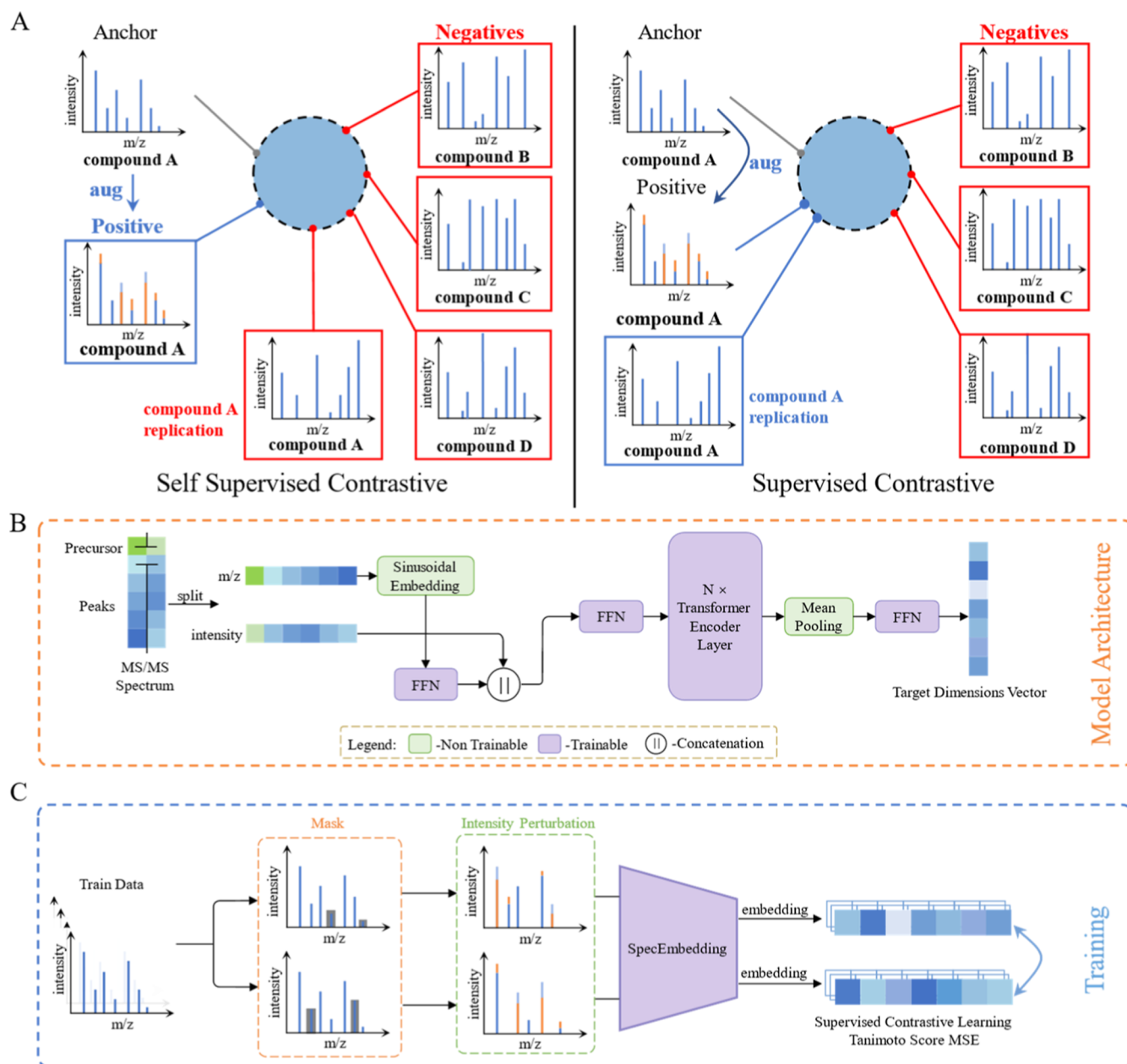


Figure 1. Illustration of contrastive learning, model architecture, and training workflow. (A) Schematic comparison between self-supervised contrastive learning and supervised contrastive learning. (B) Overall architecture of the SpecEmbedding model. (C) Training workflow of SpecEmbedding. In the intensity perturbation module, regions where the perturbed intensity is lower than the original are shown as blue translucent lines overlapping with orange lines, while regions where the perturbed intensity is higher are represented by solid blue lines overlapping with orange lines.

prediction tasks. MSBERT²¹ is built upon the BERT²² architecture, applying a masking strategy to generate positive samples and jointly optimizing model parameters through masked peak prediction and contrastive learning. In the subsequent evaluations, we systematically compare our method against the aforementioned approaches.

It is worth noting that during the initial vectorization stage, most existing methods employ peak binning²³ along the m/z dimension. For instance, Spec2Vec and MSBERT both use a bin size of 0.01 Da, while MS2DeepScore adopts a bin size of 0.1 Da. The choice of bin size typically relies on domain-specific prior knowledge: too few bins may result in insufficient resolution, whereas too many bins can lead to a sparse peak space, reducing the proportion of effective peaks and ultimately

impairing the model's generalization performance. In addition, during the self-supervised contrastive learning process,^{24,25} such as in MSBERT, positive pairs are usually generated by applying two different masking augmentations to the same spectrum, while the remaining spectra in the batch are treated as negative samples (as illustrated on the left side of Figure 1A). However, this approach overlooks the intrinsic relationships among replication spectra originating from the same compound. Since replication spectra derive from identical molecules, they should naturally be regarded as positive samples belonging to the same class.

In this study, we propose a novel method based on supervised contrastive learning (SCL),²⁶ named SpecEmbedding, aiming to achieve a more reasonable embedding of mass

spectrometry data in the vector space. SpecEmbedding trains the model to bring replication spectra originating from the same compound closer together in the feature space, while effectively separating spectra from different compounds, thereby improving compound identification accuracy. Unlike self-supervised contrastive learning approaches, which typically consider only different augmented versions of the same spectrum as positive pairs, when constructing positive pairs, SpecEmbedding incorporates not only different augmentations of the same spectrum but also replication spectra from the same compound (as shown on the right side of Figure 1A). This strategy allows the model to better leverage existing label information and learn more discriminative feature representations. Moreover, to further enhance the model's ability to perceive structural similarity, we introduce a mean squared error (MSE) weighting mechanism based on structural similarity on top of the supervised contrastive loss, forming a joint loss function (see the Method section for details). This design encourages the model to distinguish between different compounds while simultaneously capturing potential sub-structure similarities, thereby improving its generalization performance in complex sample analyses.

Data Set and Method. Data Set. In this study, the GNPS¹ data set was used for model training and testing, while the MassBank of North America (MoNA) and MTBLS1572²⁷ data sets were used for evaluation. The MSBERT team preliminarily processed the GNPS data set by applying filters for ionization mode, m/z range, minimum number of peaks, precursor m/z range, and peak intensity normalization. Additionally, they categorized the spectra according to instrument type (Orbitrap, QTOF, and Other). However, we identified a small number of entries with missing metadata or formatting errors (e.g., incorrect SMILES strings), which were subsequently removed from the data set.

In the Orbitrap data set, we divided the data into a training set and a test set (Orbitrap train set and Orbitrap test set) at the compound level, ensuring no overlap between the two. These sets were further divided into a query set and a reference set. If a compound has two or more spectra, one spectrum is randomly selected as the query set, and the remaining spectra are assigned to the reference set. If only one spectrum is available, it is placed in the reference set. We applied a similar approach to the QTOF and Other data sets when creating the query and reference sets. To eliminate random factors, we performed 10 random splits of the Orbitrap test set, QTOF, and other data sets, and the final results were reported as the average and variance of each split. The training data consisted of the Orbitrap train reference set, consistent with MSBERT. The final data distribution is shown in Table S1.

For the MoNA data set, we divided it into two subsets based on instrument type: Orbitrap and QTOF. Spectra that shared compounds with the according GNPS data set were placed into the common set, while other spectra were placed into the unique set. We used the common set for compound identification tasks and the unique set for the analogous compound searching task. Table S2 and Figure S1 illustrate the data distribution.

The MTBLS1572 data set contains mixed samples of 134 endogenous metabolites, with spectra obtained using a QTOF instrument and DDA fragmentation mode. The MSBERT team filtered precursor ions and retention times (RT), ultimately obtaining spectra for 21 compounds, which were

then divided into a query set and a reference set to evaluate the model's identification capability in mixed samples.

METHOD

The model structure is shown in Figure 1B. We only analyze the top 100 peaks ranked by intensity; if the number of peaks is fewer than 100, we pad it with 0. The spectrum is divided into two parts: m/z and intensity. To incorporate information from precursor, we place the m/z value of the precursor m/z before the peak m/z values and assign a constant intensity value of 2 to the corresponding position, so it can be distinguished from the peaks. Sinusoidal embedding^{28–31} has become a widely used method for m/z embedding in the mass spectrometry field, and we also use this encoding method, as shown in eq 1. Each m/z value is encoded as a d -dimensional vector, forming a sequence vector. After passing through a feedforward neural network (FFN) layer, the sequence is concatenated with intensity information and then passed through another FFN layer to obtain the encoded vector for the entire spectrum. Finally, the spectrum is transformed into a target-dimensional vector representation using a transformer encoder,³² mean pooling, and FFN, which can be used to calculate cosine similarity to evaluate the similarity between two spectra.

$$\begin{aligned} \text{SE}(m/z, 2i; d) &= \sin \left(2\pi \left[\lambda_{\min} \left(\frac{\lambda_{\max}}{\lambda_{\min}} \right)^{2i/(d-2)} \right]^{-1} m/z \right) \\ \text{SE}(m/z, 2i+1; d) &= \cos \left(2\pi \left[\lambda_{\min} \left(\frac{\lambda_{\max}}{\lambda_{\min}} \right)^{2i/(d-2)} \right]^{-1} m/z \right) \end{aligned} \quad (1)$$

where $\lambda_{\min}$ and $\lambda_{\max}$ represent the minimum and maximum wavelengths (in Daltons) used in the sinusoidal and cosine encoding functions. These wavelengths define the range of frequencies applied across the embedding dimensions, allowing the model to capture both fine-grained and coarse-grained variations in the input m/z values. The d denotes the embedding dimension, and we set it to 512. Given that the analysis scale is 10^{-3} Da and the maximum m/z value is 1000, we set $\lambda_{\min}$ to 10^{-3} and $\lambda_{\max}$ to 10^3 . This wavelength range ensures that the sinusoidal encoding can distinguish small differences in low-mass regions while preserving global trends over the full m/z range.

According to the compounds and their corresponding replication spectrum sequences, all data is organized with the compound as the key and the replication spectrum sequence as the value. Then, the data is split into a training set and a validation set at an 8:2 ratio, ensuring that there is no overlap of compounds between the two sets. The training set is used for model training, while the validation set is used for selecting the best model. During the training phase, inspired by MS2DeepScore,¹⁶ we adopted a similar data augmentation method, including randomly masking part of the peaks and shifting intensities proportionally (as shown in Figure 1C). Each data augmentation operation (masking positions and intensity shift values) is different, and the augmented data is also treated as a positive sample. Additionally, to fully explore the inherent relationships between replication spectra of the same compound, we introduced a probabilistic operation: with probability p , the spectrum undergoes two rounds of data

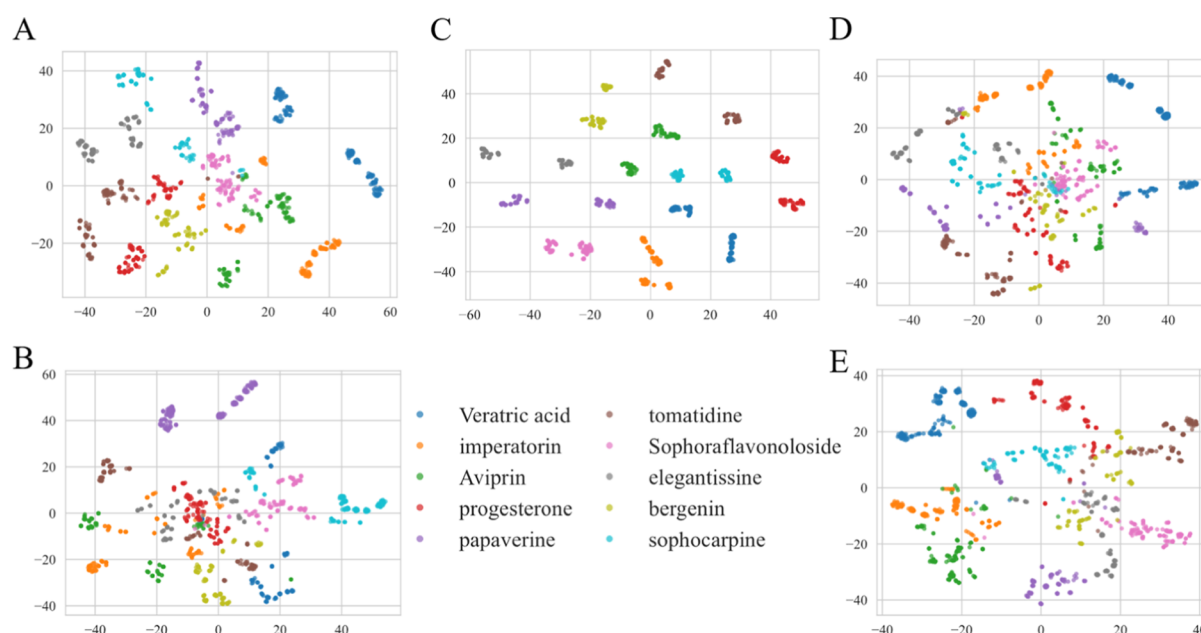


Figure 2. t-SNE visualization results of the VSET1 data set, where different colors represent different compounds. (A) MSBert. (B) DreaMS. (C) SpecEmbedding. (D) Spec2Vec. (E) MS2DeepScore.

augmentation, and with probability $1-p$, another spectrum is randomly selected from the compound's replication spectrum set. The detailed process is shown in Text S1. Whether it is the original spectrum or the augmented spectrum, if they come from the same compound, they are treated as positive samples, while the rest are treated as negative samples. To simultaneously learn molecular structure information, we calculate the similarity from the dot product of the embedding vectors and the corresponding compound's Tanimoto Score using mean square error (MSE), where the molecular fingerprint of the compound is generated using the daylight³³ algorithm based on the SMILES sequence, with a fingerprint length of 2048 bits. The final loss function is the weighted sum of both, as shown in eq 2. At the end of each training epoch, we calculate the loss on the validation set and finally save the model with the lowest loss value.

$$\begin{aligned}\mathcal{L}_{\text{out}}^{\text{sup}} &= \sum_{i \in I} \frac{-1}{|P(i)|} \sum_{p \in P(i)} \log \frac{\exp(z_i \cdot z_p / \tau)}{\sum_{a \in A(i)} \exp(z_i \cdot z_a / \tau)} \\ \mathcal{L}_{\text{out}}^{\text{t}} &= \frac{1}{N^2} \sum_{i=1}^N \sum_{j=1}^N (z_i \cdot z_j - s_{ij})^2 \\ \mathcal{L}_{\text{out}}^{\text{all}} &= \mathcal{L}_{\text{out}}^{\text{sup}} + \alpha \mathcal{L}_{\text{out}}^{\text{t}}\end{aligned}\quad (2)$$

where $P(i)$ represents the data set with the same label as sample i (excluding i itself), $A(i)$ is the set of all samples except i , and z_i is the embedding vector of sample i after being processed by the model. τ is the temperature parameter used to adjust the smoothness of the distribution. $N = n_{\text{views}} \times \text{batchsize}$, where n_{views} is the number of data augmentations and batchsize is the batch size during one training step. s_{ij} represents the Tanimoto Score between sample i and sample j .

■ RESULT AND DISCUSSION

Embedding Vector Visualization. To evaluate the rationality of the embeddings from each method, this section

visualizes the mass spectrometry spectrum embeddings using the t-SNE³⁴ algorithm. Specifically, the t-SNE algorithm is used to reduce the high-dimensional embedding vectors to a two-dimensional space, and a scatter plot is generated to display the distribution. The parameters used in the t-SNE dimensionality reduction follow the default settings in the Scikit-learn package.³⁵

The visualization experiments are based on two data sets. The first data set (VSET1) is randomly selected by the MSBert team from the GNPS Orbitrap train reference set, consisting of 10 compounds with more than 100 replication spectra. The second data set (VSET2) is selected from the GNPS Orbitrap test reference set, consisting of the top 10 compounds with the most replication spectra. The visualizations of VSET1 and VSET2 are shown in Figures 2 and S2, respectively. The results show that SpecEmbedding is capable of tightly clustering spectra from the same compound, while forming clear separation boundaries between spectra from different compounds. In contrast, the other four methods exhibit noticeable overlap between different compounds. These results indicate that SpecEmbedding has a significant advantage in distinguishing spectra, providing a solid foundation for subsequent compound identification tasks.

Compound Identification. In the compound identification task, the model first processes each spectrum to obtain an embedding vector. For a query set containing M spectra and a reference set containing N spectra, an $M \times N$ score matrix can be calculated. To evaluate the model's identification performance, the Top-k method is considered to compute the hit ratio, i.e., hit@k. Specifically, for each spectrum in the query set, if the corresponding Top-k candidate set contains the reference spectrum of the same compound, it is considered a hit. The final hit ratio is defined as the proportion of hits in the query set relative to the total number of samples in the query set, as shown in eq 3.

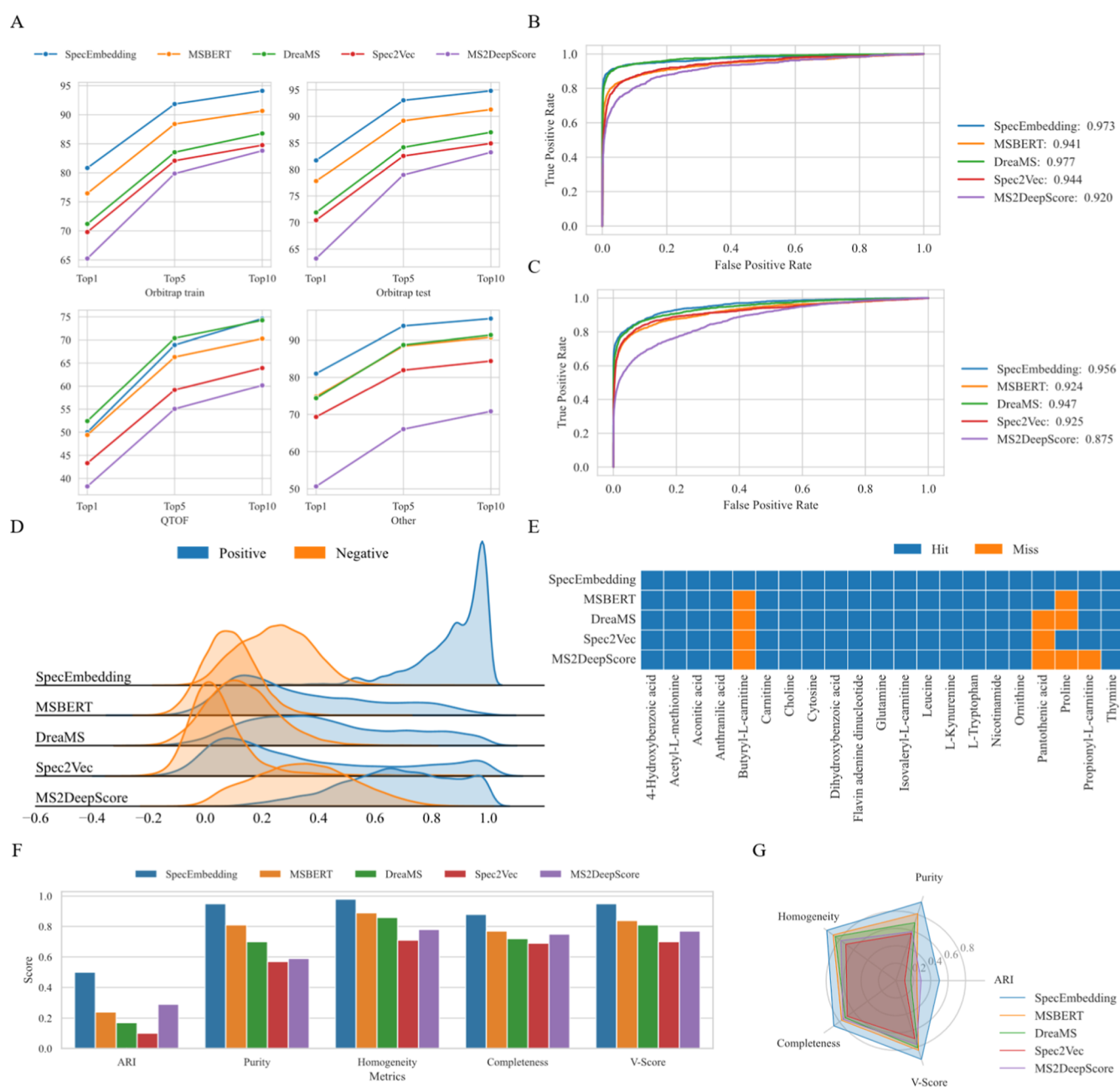


Figure 3. Results of compound identification and clustering analysis. (A) Comparison of the average hit@1, hit@5, and hit@10 scores across different subsets of the GNPS data set. (B) ROC curve for the compound identification task on the GNPS QTOF data set. (C) ROC curve for the compound identification task on the GNPS Other data set. (D) Distribution of embedding similarity scores for positive and negative sample pairs in the GNPS Orbitrap test data set. (E) Hit@1 identification results for 21 compounds in the MTBLS1572 data set. (F) Bar plot of clustering performance on the CSET1 data set. (G) Radar chart of clustering performance on the CSET2 data set.

$$\text{hit@k} = \frac{\sum_{i=1}^M I(L_i \in C_i)}{M}$$

$$I(L_i \in C_i) = \begin{cases} 0 & \text{if } L_i \notin C_i \\ 1 & \text{if } L_i \in C_i \end{cases} \quad (3)$$

where L_i represents the true compound label of the i -th spectrum in the query set, and C_i represents the set of candidate compounds ranked in the Top- k in the reference set for the i -th spectrum in the query set.

We use the hit@1, hit@5, and hit@10 metrics to evaluate the performance of each method in the compound

identification task. Table S3 and Figure 3A show the compound identification hit ratio results of each method on the GNPS data set. When testing the Orbitrap instrument subset, we combined the GNPS Orbitrap train reference set with the Orbitrap test reference set to construct a larger reference library. This setup aims to introduce additional interference information into the reference library to examine whether the model can still accurately identify the target compound corresponding to the query spectrum in the presence of substantial nontarget compound interference. It is important to note that the compounds in the training set and test set do not overlap, so theoretically, no misidentifications from training set compounds should occur during testing. For

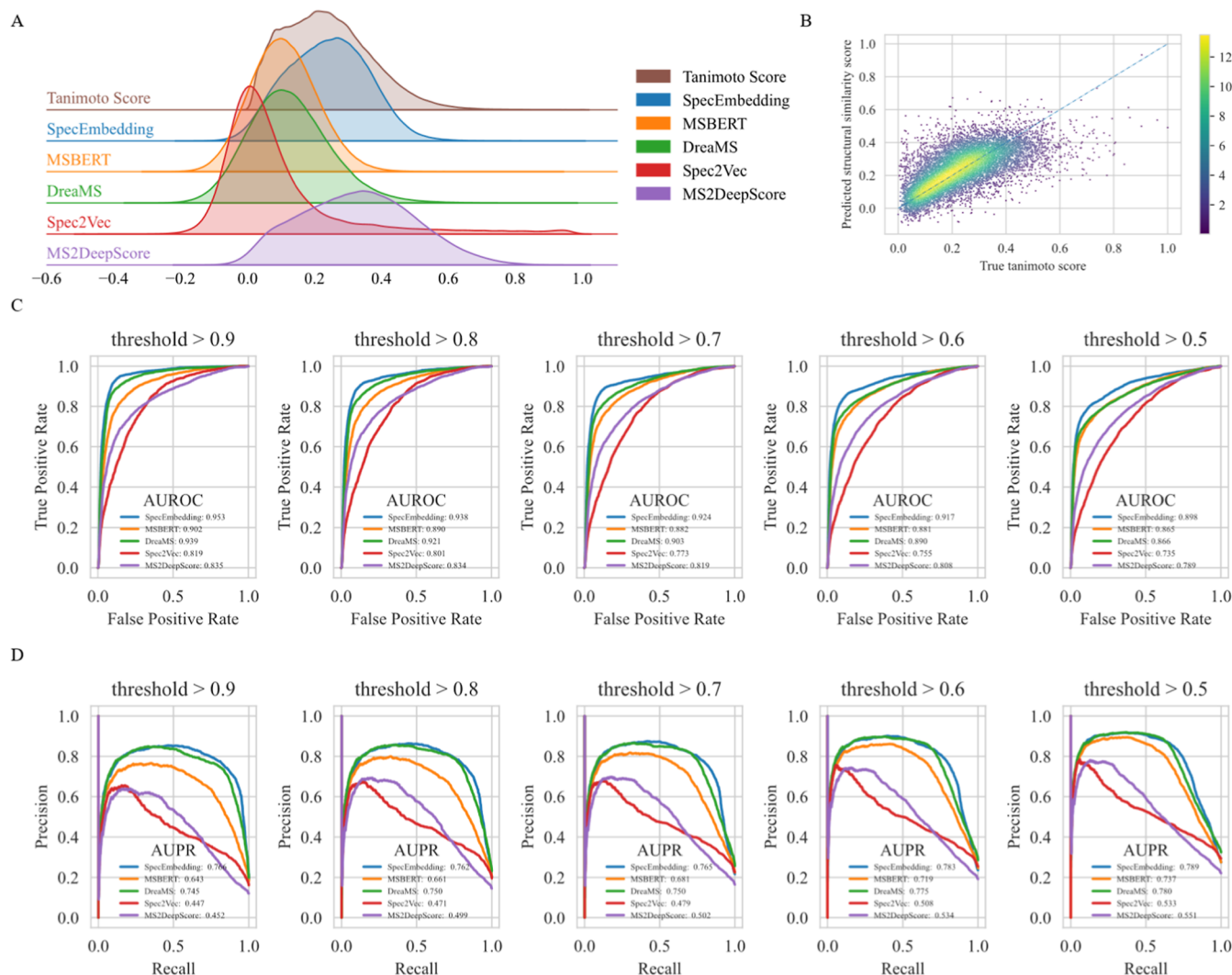


Figure 4. Relationship between embedding vector similarity and structural similarity. (A) Distribution of embedding vector similarity and true structural similarity (Tanimoto Score) on the GNPS Orbitrap test query set. (B) Kernel density scatter plot of predicted similarity and true structural similarity. (C) ROC curve of similar compound retrieval task on the MoNA Orbitrap data set at different thresholds. (D) PR curve of similar compound retrieval task on the MoNA Orbitrap data set at different thresholds.

the QTOF and Other subsets of the GNPS data set, we only use the query set to match the reference set then compute similarity scores, and evaluate the hit@k metrics based on these scores. From the results in Table S3, we can see that except for being slightly lower than DreaMS on the QTOF set, SpecEmbedding outperforms the other methods on all other data sets. Specifically, on the Orbitrap train set, Orbitrap test set, and Other set, SpecEmbedding's hit@1 average is 4.39%, 3.92%, and 6.21% higher than the current best method, MSBert. Further analysis of the performance on the QTOF data set, which is slightly lower than DreaMS, reveals that, according to the original paper of DreaMS, it used a large amount of QTOF data during pretraining. Therefore, the model already possesses prior knowledge for this instrument type. In contrast, other methods, including SpecEmbedding, only used Orbitrap data during training, lacking direct learning of QTOF instrument features. Despite the challenges posed by instrument differences, SpecEmbedding still surpasses DreaMS on the hit@10 metric, demonstrating excellent cross-instrument generalization ability.

To further evaluate each method's ability to discriminate between positive and negative sample pairs (where a positive pair corresponds to spectra from the same compound and a negative pair corresponds to spectra from different compounds), we randomly sampled 10 million sample pairs, computed the similarity scores predicted by each method, and plotted the corresponding ROC curves. The results for QTOF and Other instrument data are shown in Figure 3B,C, respectively. The legend in the figures marks the AUROC (Area Under the ROC Curve) values for each method. It can be observed that SpecEmbedding performs similarly to the best-performing DreaMS method on both data sets. Additionally, on the Orbitrap test set, we obtained similarity scores for all positive sample pairs and an equal number of randomly selected negative sample pairs. The score distribution is shown in Figure 3D. From the figure, it is intuitively evident that SpecEmbedding excels in distinguishing between positive and negative sample pairs, and compared to other methods, the overlap between positive and negative sample pairs in the kernel density plot is notably smaller. This indicates that SpecEmbedding can significantly differentiate between spectra

from the same and different compounds, effectively reducing the intraclass distance and increasing the interclass distance, thereby greatly improving the accuracy of compound identification.

When performing the compound identification task on the MoNA data set, we matched the Orbitrap common set and QTOF common set with the corresponding instrument data from GNPS, and the hit ratio results are shown in Table S4. On the Orbitrap instrument, SpecEmbedding outperformed the best method, DreaMS, on the hit@1 metric, improving by 3.79%.

On the MTBLS1572 data set, the authors of MSBERT constructed query and reference sets for 21 compounds, achieving a hit@1 ratio of 100%. SpecEmbedding also achieved this result. Interestingly, when we swapped the query and reference sets (i.e., the query set as the reference set and the reference set as the query set), MSBERT failed to identify some compounds, while SpecEmbedding maintained a hit@1 of 100%. The identification results for the 21 compounds under the swapped scenario are shown in Figure 3E. To further analyze this phenomenon, we plotted in Figure S4 the spectra that were missed in the query set and incorrectly identified in the reference set by other methods, along with the spectra that should have been identified theoretically. From this, we can see that all four other methods confused Butyryl-L-carnitine with Isovaleryl-L-carnitine, while Figure S5 shows the scores given by the five methods. By comparing their structures, we observe that the similar molecular structures of the two compounds caused other methods to make errors in scoring, whereas our method was able to effectively distinguish between them, further proving the model's exceptional recognition capability in compound identification.

Relationship between Embedding Vectors and Structural Similarity. In this section, we investigate the relationship between the similarity of the model-generated embedding vectors and the molecular structural similarity. We used the GNPS Orbitrap test query data set, processed each spectrum through the model to obtain its embedding vector, and computed the similarity scores between the embeddings to form a score matrix. To eliminate the interference from self-similarity, we masked the diagonal elements of the matrix. We then compared the similarity scores at the unmasked positions with the corresponding true structural similarity scores and plotted the density distribution, as shown in Figure 4A. The results indicate that the similarity distribution produced by SpecEmbedding is more consistent with the true structural similarity compared to other methods. Additionally, we randomly sampled 10,000 score pairs and plotted a kernel density scatter plot, as shown in Figure 4B. The plot shows that the predicted similarities maintain a good correlation with the true structural similarities. These findings suggest that the embeddings produced by SpecEmbedding form a more reasonable representation in the vector space compared to other methods.

External Validation on Curated Spectral Libraries. To further validate the generalizability and chemical robustness of SpecEmbedding beyond GNPS, we conducted external evaluation experiments using two curated MS/MS spectral repositories: MassBank³ and MassSpecGym.³⁶ We applied the same preprocessing protocol as described for GNPS, including instrument filtering (Orbitrap and QTOF), *m/z* range normalization, and intensity scaling. Importantly, to prevent data leakage and ensure a rigorous out-of-distribution

evaluation, we excluded all compounds that appeared in the GNPS training set. For each repository, we constructed three evaluation subsets: Orbitrap-only, QTOF-only, and a merged All-instrument data set. To simulate realistic retrieval scenarios, each subset was subjected to 10 independent random splits at the compound level into nonoverlapping query and reference sets. No compound appeared in both sets within a split. Each evaluation computed the hit@1, hit@5, and hit@10 metrics based on embedding similarity between query and reference spectra.

Table S5 summarizes the identification performance. On MassBank, SpecEmbedding achieved 89.47% hit@1 accuracy on the Orbitrap subset, 85.93% on QTOF, and 83.54% on the All-instrument setting. On MassSpecGym, we observed similarly strong performance, with 87.03% hit@1 on Orbitrap, 77.20% on QTOF, and 86.77% on All. Standard deviations across splits remained low, indicating stability. These results confirm that SpecEmbedding trained solely on GNPS can generalize effectively to independently curated data sets, preserving chemically meaningful structure in the embedding space.

Analogous Compound Searching. When the target compound is not present in the reference set, the Top-1 compound retrieved based on similarity scores is typically the one with the most similar molecular structure. In this section, we investigate the performance of different methods on the similar compound retrieval task. We use the MoNA Orbitrap common set as the reference set and the MoNA Orbitrap unique set as the query set, ensuring that there is no overlap in compounds between the two sets. In this task, the compound with the Top-1 embedding similarity is considered the most structurally similar compound, and we apply different thresholds (ranging from 0.9 to 0.5) to label samples. If the structural similarity of a compound exceeds the threshold, it is regarded as a positive sample; otherwise, it is considered negative. The embedding similarity is then used as the predicted probability to plot the ROC and PR curves. The corresponding curves are shown in Figure 4C,D. As illustrated, SpecEmbedding achieves higher AUC values than DreaMS in both ROC and PR curves, indicating that SpecEmbedding is better at recognizing highly similar molecular structures from spectra and embedding them more closely in the vector space.

Spectral Clustering. Spectral clustering groups multiple similar spectra into the same category to support downstream analyses,⁴ and is commonly used in untargeted metabolomics studies and spectral annotation tasks.^{10,37} In this section, we investigate the clustering performance based on the embedding vectors generated by different methods. We applied the Birch³⁸ clustering algorithm to the embeddings and evaluated clustering quality using metrics such as Adjusted Rand Index (ARI),³⁹ purity, homogeneity, completeness, and V-score⁴⁰ (refer to Text S2 for detailed definitions). The clustering evaluation experiments were conducted on two data sets. We constructed the first data set (CSET1) by randomly selecting spectra corresponding to 1000 compounds from the GNPS Orbitrap training reference set, following the setup of the MSBERT study. We built the second data set (CSET2) using the GNPS Orbitrap test reference set, including spectra for all 2043 compounds. Figures 3F,G, and S3 illustrate the clustering evaluation results on the CSET1 and CSET2 data sets, respectively. The results show that SpecEmbedding outperformed the best baseline method, MSBERT, across all evaluation metrics on both data sets, demonstrating superior

clustering performance. This not only indicates that our method can effectively group similar spectra together but also indirectly validates the strong performance of SpecEmbedding in compound identification tasks.

Model Training. For the Spec2Vec model, we trained corresponding models based on three types of instruments: Orbitrap, QTOF, and Other, due to its built-in missing peak threshold. The MS2DeepScore model was trained according to the parameters set in its original paper. The DreaMS and MSBERT models directly utilized the pretrained models provided by the authors. The SpecEmbedding model was built using PyTorch,⁴¹ with the AdamW⁴² optimizer. To prevent excessively large gradients during training, we employed gradient clipping⁴³ with a clipping threshold of 0.5. To ensure reproducibility, the random seed was fixed at 42 during training. Considering that hyperparameter settings significantly impact model performance, we used the Optuna⁴⁴ framework for hyperparameter search, conducting 100 experiments (trials) during the search process. The specific hyperparameters to be searched, the search space, and the final adopted parameter values are detailed in Table S6. In each trial, after setting the hyperparameters, the model was trained, and the model with the lowest loss on the validation set was selected. Hit@1 was computed on both the GNPS Orbitrap train reference set and the GNPS Orbitrap train query set as the evaluation metric to determine the performance of each trial. The distribution of hit@1 for the 100 trials is shown in Figure S6. Additionally, to further validate the stability of the model's performance and avoid the randomness of a single random seed, we tested the performance of models trained with 10 different random seeds (42, 66, 88, 590, 666, 888, 3306, 4402, 6603, 9999). Hit@1 mean and variance values were computed on the GNPS data sets, as shown in Figure S7. These experimental settings facilitate the replication of our work by other research teams.

Ablation Study. In this section, we explore the impact of loss function design and data augmentation modules on model performance. For the selection of loss functions, we tested the performance of models trained using only supervised contrastive learning loss, only Tanimoto score mean squared error (MSE), and a weighted combination of both. The results are shown in Table S7. As seen in the table, the model using only Tanimoto Score MSE performed the worst, while the model using the combination showed the best performance. This suggests that introducing molecular structural information, in addition to learning the relationships between similar and dissimilar samples, helps further improve the model's performance. Regarding the data augmentation module, we tested the model performance with and without enabling the data augmentation strategy, as shown in Table S8. The model with data augmentation performed better across all metrics, indicating that operations like peak masking and intensity perturbation can effectively improve the model's generalization ability, allowing it to maintain high performance across different data spaces.

Failure Mode Analysis. To better understand the limitations of the proposed model, we systematically examined misidentification cases across data sets. The analysis revealed that the majority of errors were attributable to differing adduct annotations and the presence of highly similar substructures in the reference library. Specifically, in the Orbitrap data set, 26% of misidentifications involved high structural similarity and 55% involved adduct variation, with 9% meeting both criteria.

In the QTOF data set, these proportions were 43% and 68%, respectively, with 21% of errors involving both factors.

In addition to adduct-related errors, we also investigated the influence of in-source fragments (ISFs) on model performance. ISF spectra retaining their original precursor m/z exhibited substantially reduced similarity to their corresponding normal spectra, whereas correcting the precursor m/z restored similarity to a level comparable with replicate spectra, illustrated in Figure S9. This demonstrates that ISFs introduce notable perturbations to the embedding space and highlights the model's sensitivity to precursor ion information.

CONCLUSION

In this study, we propose SpecEmbedding, a mass spectrometry spectrum embedding method trained using a supervised contrastive learning strategy. Experimental results show that SpecEmbedding can more effectively embed spectra into the vector space, thereby significantly improving the accuracy of spectrum similarity calculation. Compared to existing methods such as MSBERT, DreaMS, Spec2Vec, and MS2DeepScore, it demonstrates superior performance in compound identification and similar compound retrieval tasks. Even when trained on data from a single instrument, it retains strong performance when transferred to other instrument platforms for testing. Additionally, in the spectrum clustering task, it also achieved significant advantages, laying a solid foundation for subsequent downstream analyses. Looking forward, future research could further enhance the robustness and interpretability of the framework by incorporating simulated in-source fragment (ISF) spectra, accounting for adduct variations, and exploring structure-aware similarity measures. These extensions would help mitigate common sources of spectral artifacts and systematic misidentifications, thereby broadening the applicability of SpecEmbedding to more diverse metabolomics and cheminformatics tasks.

ASSOCIATED CONTENT

Data Availability Statement

All training and testing data sets, as well as the trained model, have been uploaded to Figshare and are publicly accessible through the identifier [10.6084/m9.figshare.28876751.v2](https://figshare.com/figures/data/10.6084/m9.figshare.28876751.v2).

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.analchem.5c02655>.

Randomized spectral data generator(Text S1); cluster metric description(Text S2); GNPS data distribution after cleaning(Table S1); MoNA data distribution after cleaning(Table S2); hit@k results on GNPS data set; hit@k results on MoNA data set (Table S4); hit@k results on external data sets(Table S5); hyperparameter search and selected values and top1 hit ratio distribution across seeds on GNPS data sets(Table S6); hit ratio with different loss functions(Table S7); hit ratio with vs without data augmentation(Table S8); overlap between MoNA and GNPS data sets(Figure S1); t-SNE visualization of the VSET2 data set(Figure S2); clustering performance metrics on CSET2 data set(Figure S3); spectra of compounds not identified by four other methods on the MTBLS1572 data set(Figure S4); top1 scores and corresponding molecular structure of Butyryl-L-carnitine identification by five methods(Figure S5); scatter plot of search evaluation metrics from 100

hyperparameter tuning experiments (Figure S6); top1 hit ratio distribution across seeds on GNPS data sets (Figure S7); ISF analysis (Figure S8); two representative misidentification cases likely caused by highly similar substructures in the library (Figure S9) (PDF)

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Author Contributions

[†]P.X. and H.X. contributed equally to this work. P.X. and H.X. contributed equally. H.Z. supervised all aspects of the study. All authors were involved in the design of this work.

Notes

Code Available The source code for this work is publicly available at <https://github.com/sword-nan/SpecEmbedding>. The authors declare no competing financial interest.

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